Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Dataset Overview

3,900

Total Purchases

Comprehensive transaction data

18

Data Columns

Customer demographics & behavior

37

Missing Values

Only in Review Rating column

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Data Preparation Process

Data Loading

Imported dataset using pandas

Initial Exploration

Structure check and summary statistics

Missing Data Handling

Imputed Review Rating with median values

Feature Engineering

Created age groups and purchase frequency

Database Integration

Connected to PostgreSQL for analysis

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Revenue by Gender Analysis

Key Insight

Female customers generate slightly higher total revenue than male customers

Strategic Implication

Gender-based marketing strategies could optimize revenue streams

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High-Value Discount Users

Customers using discounts while spending above average

Smart Shoppers

High spenders who maximize value with discounts

Strategic Opportunity

Target premium customers with exclusive offers

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Top-Rated Products

Blouse

5 stars — Highest customer satisfaction

Dress

5 stars — Consistently excellent reviews

Shirt

4 stars — Strong customer approval

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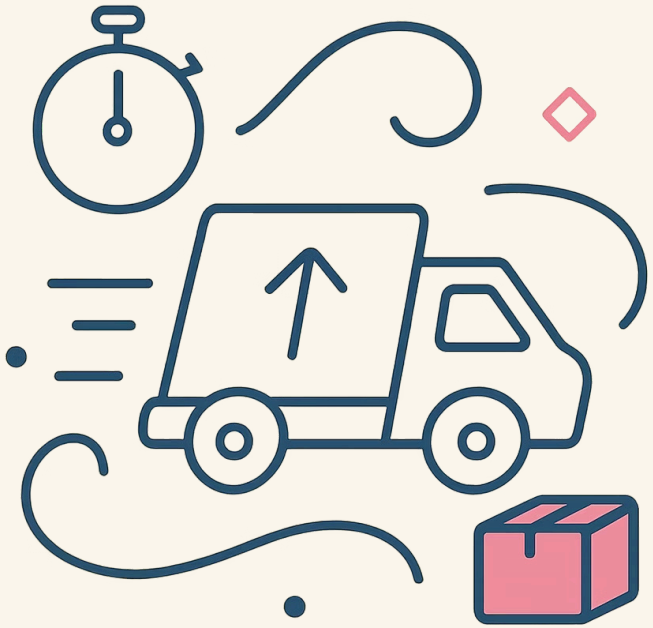

Shipping Preferences Impact

Express Shipping

$65 Average purchase amount

Standard Shipping

$58 Average purchase amount

Express shipping customers spend **12% more** per transaction

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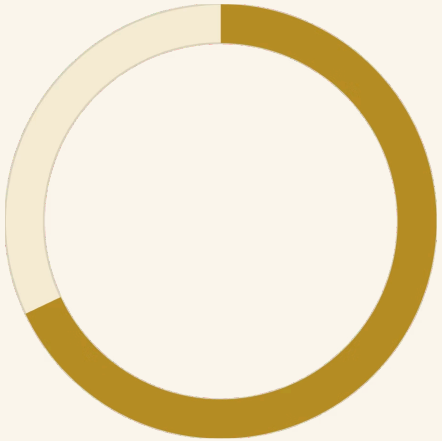
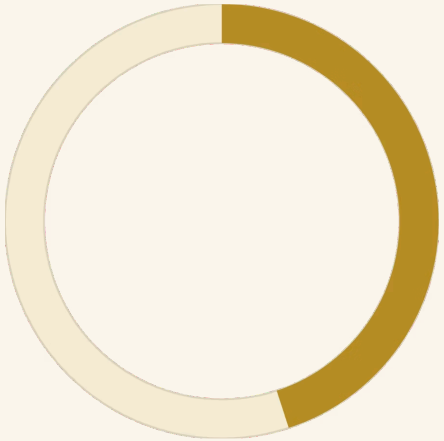
Subscription Impact



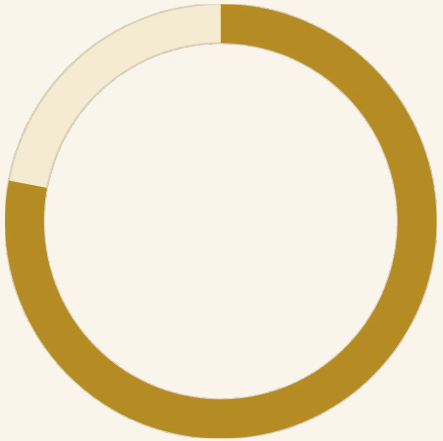
Higher Spend

Subscribers vs non-subscribers



Revenue Share

From subscription customers



Loyalty Rate

Repeat purchase frequency

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Customer Segmentation

Loyal

15% — High value customers

Returning

35% — Regular shoppers

New

50% — First-time buyers

Focus on converting New to Returning, Returning to Loyal

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Loyalty Programs

Reward repeat buyers to increase retention

Targeted Marketing

Focus on high-revenue segments and express users

Product Positioning

Highlight top-rated products in campaigns

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